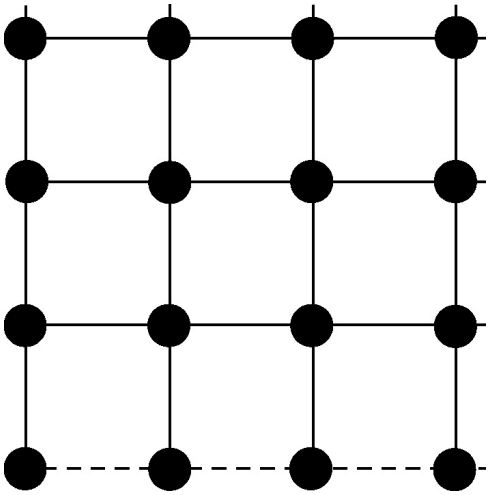


Figure 7.6 A grid approximating physical spacetime.

In quantum gravity, something beautiful occurs: the two methods of making calculations become one and the same. The spacetime foam represented in figure 7.2, used to compute a physical process in quantum gravity, may be interpreted *either* as a Feynman diagram *or* as a lattice calculation.[fn43] Therefore, the two calculation techniques used for the standard model turn out to be particular cases of a common technique: summing over the *spinfoams* of quantum gravity.

Earlier, I set out Einstein's equations. Again, I can't resist including here the complete collection of the equations of loop theory, even if the reader will obviously not be able to decipher them – not before undertaking the study of a good deal of mathematics. Someone once claimed that a theory isn't credible if its equations cannot be summarized on a T-shirt. Here is that T-shirt for loop quantum gravity (figure 7.7).